

Detailed B2B Commercial & Technical Proposal

Librika Smart Library & Learning Management Ecosystem (LMS/ILMS)

1. Executive Proposal & Project Context

This proposal defines the complete scope, technical deliverables, and commercial pricing model for the implementation of the Librika platform. Librika transforms typical school library portals into advanced learning networks by integrating high-speed automated book cataloging (VLM scanner), gamified leaderboard scores, dynamic anti-cheat quiz validations, and conceptual AI semantic search.

Project Objectives & Scope of Work

- **Librarian Automation:** Reduce book accession and cataloging time by over 95% using parallel VLM scanning.
- **Gamification Ecosystem:** Implement reading points equations tracking checkouts, returns, reviews, and comprehension tests.
- **Unified Administration:** Deliver central Super Admin dashboards managing subscription limits, schools, and moderators.
- **Mobile Native Access:** Bundle cross-platform Android and iOS Flutter apps for student bookshelves.

Deliverables Matrix & Target SLA Indicators

Deliverable Item	Specification Detail	Performance SLA Target
Web Portal Dashboard	Central Flask admin dashboard & student profile hubs with SQLite WAL mode.	99.9% uptime, <150ms server response time.
Mobile Applications	Native Android & iOS mobile wrappers supporting offline reading.	Smooth rendering, offline DB caching active.
AI VLM Scan Engine	Multi-tiered fallback scanner (VLM -> Google Books -> DDG search).	Processes 10 books in 5 seconds parallelly.
AI Quiz Comprehension	Completions quiz generator with anti-exploit reading locks.	Zero-latency generation upon book returns.
RFID Gate Integration	UHF reader interface syncing checkouts to SQLite transactions.	Scans up to 50 tags per second at exit gates.

2. Detailed Modules & Feature Breakdown

Module Name	Feature Sub-Components	Detailed Specification Details
RFID Library	Gates, Reader Sync, Tag Writer	Real-time RFID gate triggers, batch tags writer, desk checkout readers sync, antenna range calibration.
ILMS Core	Catalog, Accession, Fines	Accession register logs, circulation terms, overdue fines scheduler, Code-128 barcode image creator.
LMS Engine	Lessons, Courses, Grading	Course syllabus logs, lecture slide attachments, homework submissions portal, gradebook summaries.
Mobile Apps	Flutter wrapper, Shelf	Offline virtual bookshelf cache, push notification reminders, barcode scan-to-issue tools.
AI Chatbot	Virtual Research Assistant	Context-aware chatbot, book recommendations engine, `/api/chat` conversational portal.
AI Search	Semantic Search, Badges	Concept-based intent matching, `■ XX% Match` badges, hybrid offline fallback filters.
Recommendations	Discovery Boards	Categorized content discovery, trending/new arrivals boards, category auto-filters.
Dashboards	Analytics, Monitoring	Active checkouts stats, overdue registers, e-Library progress tracker, student streaks graphs.
Moderation	Upload Queue, Reviews	Notes & project upload reviews, review publication queues, compliance checklists.
Portals	Student & Faculty	Personal bookshelves, reading streaks tracker, upload center, assignment grades review.

3. Indian Infrastructure Recommendations

Infrastructure specifications are custom-designed for local Indian cloud providers (e.g. AWS Mumbai, Azure Pune, Nxtra, E2E Networks).

Deployment Tier	Virtual Machine Specs	Database & Storage Specs	Backup SLA	Cost (INR/Mo)
Small (5 Schools)	4 vCPU (Intel Xeon), 16 GB RAM, 200 GB NVMe SSD	PostgreSQL (Single Instance)	Daily snapshots to local S3	■6,000 – ■10,000
Medium (50 Schools)	8 vCPU, 32 GB RAM, 500 GB SSD, Load Balancer	PostgreSQL + Redis Cache, 1TB Object Storage	Daily DB backup, weekly full snap	■18,000 – ■35,000
Large (500+ Schools)	Kubernetes Cluster (3 Nodes), Multi App Servers	PostgreSQL DB Cluster + CDN, 5TB Object Storage	Real-time WAL replications, hourly DB backup	■70,000 – ■2,00,000

4. AI Infrastructure & Sizing Calculations

To optimize costs, we compare cloud completions endpoints with self-hosted physical servers in local networks.

Option	Hardware Specifications	Electrical & VRAM Calculations	Estimated Running Cost
Option 1: API	NVIDIA Nemotron / OpenAI API endpoints. No GPU server maintenance.	Zero server electrical overhead. Low setup time.	■5,000 – ■50,000/Month (Depends on token volumes)
Option 2: Local (RTX 4090)	1x NVIDIA RTX 4090 (24GB VRAM), 64GB RAM, 1TB NVMe, 850W PSU.	Runs Llama 3.3 / Qwen (FP8 quantization). Electricity: ~450W under load (■3,500/Mo).	Hardware: ■2.2L upfront. Maintenance: ■2,000/Month.
Option 3: Local (L40S)	1x NVIDIA L40S GPU (48GB VRAM), 128GB RAM, 2TB SSD, Server Chassis.	Runs full 32B model reasoning. Electricity: ~600W under load (■4,800/Mo).	Hardware: ■5.5L upfront. Maintenance: ■3,000/Month.
Option 4: Cluster (H100)	NVIDIA H100 (80GB HBM3 VRAM), 256GB RAM, 4TB SSD, Liquid Cooling.	Handles 100+ concurrent reasoning requests. Electricity: ~1200W (■9,800/Mo).	Hardware: ■18L+ upfront. Maintenance: ■15,000/Month.

5. RFID Hardware & Indian Supplier Estimates

Evaluated RFID components sourced from leading Indian B2B suppliers and OEM vendors (Zebra India, Honeywell, GAO RFID).

Hardware Item	Technical Specifications & Models	Est. Unit Price (INR)	Primary Role & Features
RFID Tags	UHF EPC Gen2 Class 1, ISO 18000-6C (Alien H3 / Monza R6)	₹8 – ₹15 / tag	Adhered to book covers; includes anti-peel adhesive and 96-bit EPC memory.
RFID Reader	UHF Long-Range Fixed Reader (Zebra FX9600 / Invengo), 4-Port	₹75,000 – ₹1,20,000	Connects to antennas at issue desks and stock checking units.
RFID Security Gate	UHF Walk-through Security Gate (Zebra / Invengo), Dual Pedestal	₹1,80,000 – ₹2,50,000	Monitors outgoing books at exit doors. Triggers alarms for unpaid assets.
Barcode Scanner	2D Cordless Area-Imaging Scanner (Honeywell Voyager 1472g)	₹4,500 – ₹8,000	Acts as cataloging backup scanner; parses ISBN barcodes and student IDs.
Label Printer	Desktop Thermal Barcode Printer (Zebra GT800 / TSC TE244)	₹14,000 – ₹22,000	Prints spine labels and barcode tags on demand.
Self Check Kiosk	Interactive Self-Issue Kiosk (21" screen, RFID pad, printer)	₹1,50,000 – ₹2,50,000	Allows students to perform self-checkout and return books autonomously.

6. Software Development Cost Breakdown (Indian Rates)

Role Title	Monthly Rate (INR)	Duration (Months)	Subtotal Cost (INR)	Core Responsibilities
Project Manager	₹1,20,000	6 Months	₹7,20,000	SLA management, stakeholder reporting, milestones validation.
Senior Backend Developer	₹1,80,000	6 Months	₹10,80,000	Flask routing, database optimization, SQLite WAL configs, API security.
Frontend UI Developer	₹1,20,000	5 Months	₹6,00,000	Responsive web dashboards, 3D card CSS translations, OPAC design.
Flutter Developer	₹1,30,000	4 Months	₹5,20,000	Cross-platform mobile packaging, notifications sync, virtual bookshelf.
AI / VLM Engineer	₹2,20,000	4 Months	₹8,80,000	Nemotron completions integrations, fallback accuracy engine, quiz parser.
UI/UX Designer	₹90,000	2 Months	₹1,80,000	Figma prototyping, layout grids, user onboarding wireframes.
QA & Testing Engineer	₹80,000	3 Months	₹2,40,000	Unit testing, multi-user concurrency stress tests, integration checks.
DevOps Engineer	₹1,50,000	2 Months	₹3,00,000	PostgreSQL cluster config, load balancers, backups replication S3.
Total Software Budget	-	-	₹45,20,000	Excludes standard GST (18%)

7. Project Timeline & Gantt Schedule

Phase	Timeframe	Key Milestone Deliverables	Prerequisites / Dependencies
Phase 1: Planning	Month 1 (W1-2)	Database schema definitions, API layout diagrams.	Stakeholder initial sign-off.
Phase 2: UI/UX	Month 1 (W3-4)	High-fidelity Figma wireframes, interactive checkout mockup.	Phase 1 specification sheets.
Phase 3: Backend	Months 2-3	Core routing APIs, library transaction logs, database WAL.	Schema validations.
Phase 4: Frontend	Months 3-4	Mobile wrappers, librarian dashboard views, library cards.	Backend API completion.
Phase 5: AI Engine	Month 4	VLM cover scanning APIs, quiz anti-cheat lock triggers.	Prompt templates, API key setups.
Phase 6: Testing	Month 5	Stress testing concurrent library writes, barcode scanner sync.	All code-freeze modules.
Phase 7: Deploy	Month 6 (W1-2)	Server VM instance setups, DB seeds, cloud deployment.	Successful QA run.
Phase 8: Training	Month 6 (W3-4)	Librarian dashboard walkthroughs, student manuals.	Live deployment completion.

8. B2B Commercial Quotations & Pricing Plans

Package Level	Target Institution Profile	Included Scope Items	Price Package Estimate
Basic Package	Small schools, local public libraries	Web Portal, OPAC access, Barcode scanner cataloging.	■25 – ■35 Lakhs
Professional Package	Colleges, multi-campus institutions	Web portal, Android/iOS Apps, AI scanner + semantic search, 3D billing.	■45 – ■65 Lakhs
Enterprise Package	Universities, state library systems	Full RFID integrations, AI quizzes, custom state-level dashboards.	■80 Lakhs – ■1.5 Crore+

Annual Maintenance Contract (AMC) SLA Terms

AMC Tier	Annual Rate	SLA Incident Response Times & Benefits Included
Basic	15% of License/Yr	Minor bug fixes, monthly database backups. Email-only support tickets. Severity 1 response: 24 hrs.
Premium	20% of License/Yr	Weekly full backups, system patches, minor upgrades. Phone/Slack support. Severity 1 response: 8 hrs.
Enterprise	25% of License/Yr	Hourly replicas, 99.9% uptime SLA, security audits, hardware replacement help. Severity 1 response: 2 hrs.

Payment & Implementation Milestones

- **Milestone 1:** 20% Advance payment upon signing contract and scope agreement.
- **Milestone 2:** 30% Upon completion of backend API infrastructure and database WAL configurations.
- **Milestone 3:** 30% Upon completion of UI/UX, student portals, and mobile Flutter app wrapping.
- **Milestone 4:** 10% Upon completing QA testing, hardware RFID UAT checks, and dashboard training.
- **Milestone 5:** 10% Upon live deployment and database handovers.

Proposal Authorized By: Librika Commercial Sales Team.

*Pricing excludes standard GST (18%) where applicable.